

Comprehensive Analysis of Hydrological Risk, Infrastructure Resilience, and Water Resource Management in India

Introduction: The Intersection of Hydrological Extremes and Infrastructure Liability

The vulnerability of modern infrastructure is most acutely exposed at the intersection of water and earth. Across the Indian subcontinent, cyclical monsoon patterns and rapidly evolving climate dynamics are systematically dismantling engineering paradigms that have governed infrastructure design for decades. Traditional assumptions regarding rainfall intensity, aquifer recharge, irrigation efficiency, and flood magnitudes are proving catastrophically inadequate. What was once considered a 50-year design storm now manifests with alarming regularity, overwhelming infrastructure sized to obsolete hydrological constants. The consequences are not merely academic; they translate into catastrophic pavement collapses, severe groundwater depletion, massive agricultural water losses, and the devastating failure of hydraulic structures.

This comprehensive research report systematically investigates four critical domains of water and infrastructure engineering: Storm Drainage and Flood Management, Groundwater and Hydrogeology, Irrigation Water Management, and Hydraulic Structures Engineering. By synthesizing forensic engineering audits, national groundwater assessments, hydrological modeling data, and historical case studies, this analysis moves beyond surface-level symptoms to diagnose the underlying mechanisms of infrastructure failure. The objective is to elucidate how second and third-order systemic vulnerabilities—such as the cyclical hydraulic stress in pavements, the geotechnical consequences of urban dewatering, the ecological paradox of unlined canal seepage, and the inadequacy of deterministic spillway capacities—generate profound financial and structural liabilities.

Ultimately, the paradigm of water infrastructure must shift from reactive mitigation to predictive, analytically validated resilience. Systems must be engineered not for the climate that existed when foundational codes were written, but for the non-stationary hydrological reality of the present and future.

Domain 01 of 04: Storm Drainage & Flood Management

Every monsoon season exposes an uncompromising truth: the vast majority of Indian highway networks and urban drainage systems were not designed to accommodate the rainfall intensities they are now receiving. What was once categorized as a 50-year storm event now arrives every few years, and the infrastructure sized according to these outdated assumptions

fails visibly and expensively. Drainage engineering is routinely postponed during project execution until it becomes the most expensive line item on the project in the form of post-failure rehabilitation. When drainage fails, the pavement above it is already structurally compromised; the visible surface damage simply has not yet manifested.

The Illusion of Pavement Failure: A Consequence of Subsurface Saturation

A pervasive misconception within highway engineering and infrastructure maintenance is that premature pavement deterioration is primarily a materials or load-bearing failure. In reality, forensic audits conducted by the Ministry of Road Transport & Highways (MoRTH) and the Central Road Research Institute (CRRI) consistently identify inadequate drainage as the primary root cause of premature pavement failure on national highway and expressway corridors.¹ The deterioration of highway pavement accelerates sharply the moment surface water infiltrates and reaches the sub-base layers.

The mechanics of this failure are strictly hydrological and geotechnical. At the initial stage of water infiltration, the increase in soil moisture content elevates the degree of saturation within the subgrade.⁴ This moisture influx significantly reduces the soil's stiffness and shear strength. Research indicates that the reduction in matric suction due to moisture increase reduces the effective stress within the soil skeleton, promoting a rapid volume reduction and structural compression.⁴ The Central Road Research Institute and allied studies confirm that once surface water infiltrates the sub-base, subgrade shear strength can plummet exponentially; as moisture content approaches absolute saturation, no amount of surface repair can prevent structural collapse from below.⁴ A detailed investigation into subgrade moisture revealed that without adequate soaking mitigation, each 1% increase in moisture content caused a strength change rate approximately one to seven times higher than the rate observed after prolonged soaking, indicating that rapid infiltration is immediately destructive.⁴

This chemical and physical degradation is vastly compounded by dynamic traffic loads. Historically, practitioners have viewed the separation of asphalt binder from aggregate—termed "stripping"—as a physio-chemical incompatibility.⁷ However, extensive case studies indicate that under saturated conditions, all asphalt mixes are susceptible to mechanical failure.⁷ As heavily loaded vehicles traverse saturated pavements, cyclical hydraulic stress physically scours the asphalt binder from the aggregate.⁷ A typical dense-graded Hot Mix Asphalt (HMA) wearing course may possess up to 8% air voids upon construction, but traffic loading rapidly reduces this to 4-5%, sealing the surface.⁷ When subsurface drainage is inadequate, moisture vapor from the subgrade is trapped beneath this sealed wearing course.⁷ Once this moisture condenses, rutting begins in heavily loaded lanes, channeling even more surface water into the failure zones and deepening the collapse exactly as CRRI audits document in premature National Highway failures.¹

Climate Change and the Obsolescence of

Intensity-Duration-Frequency (IDF) Curves

The fundamental inputs for highway drainage design—Intensity-Duration-Frequency (IDF) curves—are increasingly disconnected from the reality of modern weather systems. Culverts sized on outdated IDF data consistently fail during peak monsoon discharge, washing out road embankments and triggering network closures that cost multiples of what proper hydraulic design would have originally required.⁸

The scientific community recognizes the non-stationarity of extreme rainfall, a phenomenon deeply intertwined with anthropogenic climate change. The theoretical baseline for this intensification is the Clausius-Clapeyron (CC) relationship, which postulates a 7% increase in atmospheric precipitation capacity per 1 °C increase in global temperature.¹⁰ However, recent studies utilizing bootstrap methods to calculate extreme precipitation changes have revealed significant regional deviations. While inland temperate zones often adhere to the 7% CC relationship, tropical and sub-tropical environments often exhibit sub-CC relationships, indicating that generalized global scaling factors are insufficient for regional infrastructure design.¹⁰ Conversely, advanced analyses utilizing an ensemble of General Circulation Models (GCMs) across varying Representative Concentration Pathways (RCP) scenarios demonstrate a uniform increase in sub-daily precipitation intensities (15, 30, 45, 60, 120, and 180 minutes) across all return periods as climate scenarios intensify.⁸

Consequently, the reliance on static historical data constitutes a profound engineering liability. Drainage must be sized for the storms that will actually occur, not the ones that historically characterized the region.

Regulatory Frameworks and Advanced Hydraulic Modeling Solutions

To combat the systemic failure of drainage infrastructure, the Indian Roads Congress (IRC) formalized rigorous guidelines through IRC:SP:42-2014 ("Guidelines on Road Drainage"). This standard emphasizes that pavement structures must be protected from any water ingress to maintain functional efficiency, dictating the necessity of both surface and subsurface water management.¹¹ Surface drainage requires adequate camber, longitudinal gradients, and the integration of kerb and gutter systems in urban settings to rapidly channel runoff.¹¹ Subsurface drainage, arguably more critical, mandates the use of geo-filter fabrics and geotextiles to act as separation layers, preventing the choking of aggregate drains and aiding in the uniform distribution of structural loads to the subgrade.¹⁵

However, geometric design compliance is insufficient without rigorous analytical validation from first principles. Modern stormwater master planning requires the integration of site-specific hydrological analysis with 1D and 2D hydraulic modeling. Tools such as the US EPA Storm Water Management Model (SWMM) and the Hydrologic Engineering Center's River Analysis System (HEC-RAS) have become indispensable for urban townships, industrial parks, SEZ developments, and bridge crossings. SWMM is optimized for urban sewer network hydrology and evaluating Low Impact Development (LID) or Sustainable Urban Drainage Systems (SuDS), while HEC-RAS excels at modeling river dynamics, water surface profiles, and channel

sediment transport.¹⁷

The necessity of integrating these tools is perfectly illustrated by a recent hydrological assessment of Hyderabad, an urban center plagued by recurring floods. A SWMM-based model simulating a 48-hour storm event (from October 3–4, 2015) revealed that the existing urban drainage infrastructure is critically insufficient, failing to convey runoff from a design storm with a mere 2-year return period.¹⁹ The rapid urbanization of Hyderabad has vastly increased impervious surfaces while decreasing natural infiltration, overwhelming drains originally sized for historical rainfall intensities of 12–20 mm/hr.¹⁹

Critical Node	Modeled Flood Depth (meters)	System Status
Node-1	2.23	Overflowing
Node-2	3.09	Overflowing
Node-3	2.99	Overflowing
Node-4	3.31	Overflowing
Node-11	3.81	Overflowing
Node-13	5.70	Maximum Capacity Reached
Outfall	5.70	Maximum Capacity Reached

Data Source: SWMM-based dynamic runoff response study for Hyderabad Stormwater Zone.¹⁹

During peak flooding, critical nodes such as Node-13 and the main outfall reached their maximum depth capacity of 5.70 meters, forcing excess water to spill over and collect in

low-lying intersections.¹⁹ The integration of SWMM hydrology with HEC-RAS flood inundation mapping provides planners with the empirical data required to size attenuation ponds, design pumping sumps, and redesign drainage capacities to manage larger volumes of surface water.¹⁹ Furthermore, culvert and cross-drainage hydraulics must incorporate box, pipe, and arch culvert sizing using HY-8 and HEC-RAS, alongside scour depth estimation per IRC:89, ensuring riprap and gabion protection design at all outfalls to dissipate high-velocity energy. Every system must be analytically validated against CWC and BIS codes before drawings are issued for construction, yielding a 100% regulatory approval record.

Domain 02 of 04: Groundwater & Hydrogeology

India functions as the world's largest consumer of groundwater, yet its resource management strategies remain profoundly inefficient. A generation ago, borewells were routinely 100 feet deep; today, they frequently exceed 800 feet and still fail to yield sustainable supply. The core vulnerability is that India's aquifers are being spent drastically faster than monsoons can replenish them, and the vast majority of developmental projects never measure or account for the deficit they create. For developers, industrialists, and urban planners, the question is no longer whether groundwater risk exists, but whether that risk has been accurately quantified.

The Silent Crisis of Aquifer Depletion and Over-Exploitation

Groundwater deficits accumulate in silence. Unlike surface reservoir depletion, aquifer overdraft remains largely invisible until it manifests in catastrophic systemic failures: failed public supply borewells, land subsidence, and the mobilization of deep-earth contaminants.

The latest National Compilation on Dynamic Ground Water Resources of India, published by the Central Ground Water Board (CGWB) in 2024, provides a stark quantification of this silent crisis. The total annual groundwater recharge in the country was assessed at 446.90 Billion Cubic Meters (BCM), yielding an extractable resource of 406.19 BCM after allocating for natural discharge.²² With a total annual groundwater extraction volume of 245.64 BCM across all sectors, the national average stage of groundwater extraction stands at 60.47%.²²

While the macro-level national extraction average suggests a semi-stable equilibrium, granular block-level data paints a dire picture of severe regional overdraft. Out of 6,746 assessment units (Blocks, Mandals, and Talukas) evaluated nationwide, 751 units—representing 11.13% of the country—are categorized as strictly 'Over-exploited', meaning annual extraction vastly exceeds total annual recharge.²²

Groundwater Categorization (2024 Assessment)	Number of Assessment Units	Percentage of Total Units

Safe (< 60% utilization)	4,951	73.4%
Semi-critical (60-90% utilization)	711	10.5%
Critical (91-100% utilization)	206	3.05%
Over-exploited (> 100% utilization)	751	11.13%
Saline (Brackish/Saline Phreatic Aquifers)	127	1.8%

Data Source: CGWB National Compilation on Dynamic Ground Water Resources of India, 2024.²⁴

Although the percentage of over-exploited blocks has declined from 17.24% in 2017 to 11.13% in 2024—largely attributed to an 11.36 BCM increase in artificial recharge from tanks, ponds, and WCS (Water Conservation Structures)—the absolute volume of depletion in critical agricultural and urban zones like Punjab and Haryana remains utterly unsustainable.²⁴ In deeply affected north-western districts, water tables are projected to drop below 300 meters by 2039 if current extraction vectors hold. Rapid urbanization in cities like Delhi and Mumbai has converted vast tracts of natural recharge zones into impermeable concrete, disrupting the monsoon recharge mechanisms that historically replenished aquifer storage.

This depletion is irrefutably corroborated by macro-level satellite gravimetry. NASA's Gravity Recovery and Climate Experiment (GRACE) satellite data highlights that the Gangetic Basin, a vital resource for one of the world's most densely populated agricultural sectors, exhibits some of the highest aquifer depletion rates recorded globally, characterized by marked seasonal fluctuations.²⁶ Research utilizing downscaled GRACE data confirms that anthropogenic impacts—specifically intensive agricultural extraction and canal network disruptions—dominate groundwater storage decline, vastly outweighing natural climatic variability.²⁸ Extensive irrigation practices increase atmospheric humidity via evapotranspiration, paradoxically altering local precipitation conditions while simultaneously draining the subterranean reserves that sustain

them.²⁶

Construction Dewatering: Geotechnical Risks and Legal Liabilities

Beyond broad agricultural extraction, localized deep urban excavation presents acute hydrogeological risks. Construction dewatering—the continuous pumping required to temporarily lower the groundwater table to facilitate dry excavation for deep foundations, mine pits, basements, and tunnel drives—is frequently executed without rigorous hydrogeological site assessment. The localized, rapid extraction of groundwater inherently removes hydrostatic pressure from the surrounding geological matrix. This reduction in pore water pressure triggers a corresponding increase in effective stress within the soil skeleton, which initiates consolidation and subsequent differential settlement of adjacent structures.³⁰

When dewatering is improperly modeled, the liability footprint extends far beyond the construction site boundaries. The construction of underground mass transit corridors provides prime examples of this severe risk. During the execution of the Mumbai Metro Line 3, geotechnical engineers recognized that extensive dewatering to excavate underground stations in dry conditions would induce elastic shortening and massive consolidation of soil layers, severely threatening the structural stability of historically significant heritage buildings in South Mumbai.³³ To mitigate this catastrophic risk, the project utilized closed-mode Earth Pressure Balancing (EPB) Tunnel Boring Machines (TBMs) and implemented pre-excavation ground treatment (pre-grouting) to minimize water ingress, thereby strictly controlling groundwater drawdown outside the station boxes.³³ The vulnerability of such subterranean infrastructure was further underscored when an unauthorized private borewell drilling operation pierced and flooded the Metro Line 3 tunnel near CSMT, highlighting the chaotic, unmapped nature of urban groundwater exploitation.³⁵

Similarly, the Delhi Metro Phase IV expansion (Aerocity to Tuglakabad corridor) required exhaustive Archaeological Impact Assessment Studies to model the interference of bored tunnels and cut-and-cover stations with groundwater flow, ensuring that five ancient monuments around the Mehrauli Badarpur Road were not compromised by dewatering-induced ground settlement.³⁶ EIA studies for the Najafgarh-Dhansa Bus Stand corridor similarly mandated strict environmental conservation measures, capping water requirements and mandating CGWA approvals for any tube well construction.³⁷

The failure to predict and mitigate these hydrogeological interferences routinely results in severe legal and financial consequences. Industrial and mining contamination plumes can migrate through permeable geology for years before surfacing at public supply borewells; at that point, the remediation liability and regulatory consequence are both severe and expensive to dispute. In India, construction contracts are generally governed by standard forms influenced by FIDIC, the Institution of Civil Engineers (ICE), or the Indian Institute of Architects (IIA).³⁸ The allocation of risk regarding geotechnical uncertainties often leads to complex litigation.³⁸ When dewatering damages adjacent private property, owners routinely pursue construction claims against project developers, lead civil engineers, and geotechnical consultants.³² Indian courts recognize claims

for building settlement caused by construction activities; for instance, in *CS No. 383/14 (Complete Dewatering Systems Pvt. Ltd. Vs. Integral Construction Co. India)*, courts actively adjudicate damages and interest based on contractual breaches related to dewatering operations.⁴⁰ While exclusion or limitation of liability clauses are valid under Indian law, courts retain discretion to award reasonable compensation in cases of liquidated damages.⁴¹

To neutralize this liability, aquifer behavior must be characterized with analytical precision, not approximated with assumptions. Best practices demand the integration of field hydrogeological investigation—including borehole siting, lithological logging, and aquifer testing interpreted using Theis, Cooper-Jacob, and Neuman methods—with numerical groundwater modeling. Utilizing software such as MODFLOW and FEFLOW to simulate steady-state and transient flow allows engineers to accurately predict dewatering drawdown, assess aquifer recovery, and map DRASTIC vulnerability indexing for EIA chapters and CGWB licensing applications, ensuring that regulatory submissions are approved on first review.

Domain 03 of 04: Irrigation Water Management

India possesses the largest irrigated agricultural footprint globally, yet it operates with staggering systemic inefficiency. In an era defined by depleting aquifers, erratic monsoon patterns, and escalating energy costs for pumping, the profound loss of agricultural water is both a food security threat and a massive financial deficit. The fundamental failure lies not in a lack of resources, but in flawed conveyance engineering. Nearly half of every liter diverted for irrigation in India is lost before it ever reaches the field; the farmer at the tail end of the canal is not a victim of drought, but a victim of engineering that was never executed properly.

The Conveyance Efficiency Paradox: Quantifying Massive Seepage Losses

Data compiled by the Food and Agriculture Organization (FAO) Aquastat system reveals a grim reality regarding Indian agriculture: approximately 45% of the water diverted from headworks for agricultural purposes is lost in transit within conveyance systems before it ever reaches the crop root zone.⁴² Consequently, macro-irrigation schemes must be constructed at roughly double the capacity that a precisely engineered system would require to deliver the identical agricultural output.⁴⁶

A definitive hydrosatial modeling case study of the Nira Left Bank Canal (NLBC) in the Pune district of Maharashtra perfectly quantifies this debilitating loss. The NLBC is an unlined earthen canal infrastructure dating back over a century, suffering from significant operational degradation, extensive vegetation, and sediment deposition. Utilizing a hydraulic model built on a spatial platform and verified with rigorous flow-monitoring events across a 30-kilometer study area, engineers demonstrated an average water conveyance loss of 39.96%.⁴⁴

Month	Conveyance Loss (%) in NLBC
August	15.96%
September	14.99%
October	14.99%
November	13.05%
December	12.45%
January	13.05%
February	13.54%
March	14.02%
April	16.92%
May	18.95%

Data Source: Monthly flow-monitoring observations utilizing standing wave flumes at the head and 30th km of the Nira Left Bank Canal.⁴⁴

In-depth analysis of these volumetric losses indicates that subsurface seepage and deep

percolation account for an overwhelming 98.37% of the total stream loss, while evaporation accounts for a negligible 0.3%.⁴⁴ The absence of canal lining, coupled with structural cracks and breaches, allows millions of liters of water to percolate uselessly into the surrounding geology. The direct victims of this engineering failure are the "tail-end" irrigators. Deprived of adequate supply due to massive upstream transit losses, these agriculturalists face severe, chronic water scarcity that yields uneven crop production and perpetuates economic hardship.⁴⁴ The implementation of Cement Concrete (C.C.) lining could mitigate these seepage losses by an estimated 30–40%, which would theoretically allow for an increase in the intensity of irrigation beyond the current 45% of net Culturable Command Area (CCA).⁴⁴

The Ecological Disaster of Unmanaged Seepage: The IGNP Case Study

While unlined canals deprive tail-end users of water, the sheer volume of seepage lost upstream frequently generates devastating secondary ecological disasters. The Indira Gandhi Nahar Pariyojana (IGNP) in Rajasthan serves as the ultimate cautionary tale of macro-irrigation implemented without holistic drainage planning and hydrogeological foresight.

Conceived as an ambitious project to transform the vast, arid Thar desert into a fertile agricultural basin, the IGNP was allocated a share of 8.6 Million Acre Feet (MAF) from the surplus waters of the Ravi and Beas rivers, intending to irrigate an area of 19.6 lac hectares across districts like Sriganganagar, Hanumangarh, and Bikaner.⁴⁷ Initially, the command area exhibited a remarkable transformation, catalyzing a localized green revolution and heavily boosting regional cropping systems.⁴⁷

However, the native geology of the Thar desert features a dense hardpan layer situated beneath sandy, highly pervious dunes, severely limiting deep aquifer percolation. The introduction of unlined earthen field canals, saddles, and the impounding of Ghaggar floodwaters in natural depressions created steep artificial hydraulic gradients.⁴⁹ Massive, continuous seepage from the unlined conveyance network caused the groundwater table to rise at an alarming, unnatural rate.⁴⁷

As the water table approached the surface, it triggered massive and intractable waterlogging. In the intense arid heat of Rajasthan, high evaporation rates drove powerful capillary action, drawing naturally occurring deep soil salts upward through the soil profile.⁴⁸ The result was catastrophic surface soil salinization. By the late 1990s, nearly 49.6% of the monitored command area in Stage-1 was deemed highly sensitive to waterlogging, rendering formerly lush and productive agricultural land barren, rocky, and utterly uncultivable.⁴⁷ While biodrainage utilizing species like Eucalyptus and Prosopis juliflora has been suggested for reclamation, the IGNP disaster proves irrefutably that applying massive volumes of water to an ecosystem without engineered subsurface drainage or strict seepage control is agriculturally suicidal.⁵⁰

Technological Intervention: SCADA Automation and Demand-Based

Scheduling

To correct the historical inequities of canal distribution, eliminate seepage-induced ecological damage, and drastically reduce the applied irrigation water volume, modern engineering is aggressively shifting toward pressurized networks, demand-based scheduling, and deep digital automation. Pressurized irrigation systems installed without rigorous hydraulic design routinely operate outside manufacturer pressure tolerances, producing uneven application uniformity and premature emitter failure. Furthermore, calendar-based irrigation schedules apply water according to administrative convenience rather than actual crop demand, leading to simultaneous over-irrigation in cool months and deficit stress during critical peak evapo-transpiration periods.

A groundbreaking example of overcoming these inefficiencies is the modernization of the Narayanpur Left Bank Canal (NLBC) in Karnataka. To address the chronic under-delivery of water to tail-end users, project authorities implemented a Supervisory Control and Data Acquisition (SCADA) based smart digital irrigation management system under the PMKSY-AIBP framework, alongside similar ongoing projects at Gosikhurd and Lower Pedhi in Maharashtra.⁵² This initiative marked the first implementation of dam-to-farmer delivery outlet automation highly customized for Indian socio-economic conditions.⁵²

The NLBC system integrates Internet of Things (IoT) based, solar-powered, fully automated stainless steel gates specifically designed to be vandalism-proof.⁵² These advanced gates feature built-in capabilities for direct discharge measurement, upstream and downstream water level telemetry, and gate actuator positioning.⁵² Data from the physical infrastructure is fed into a centralized water audit and decision support system via wireless data communication, closely monitored by CCTV surveillance.⁵²

Coupled with an Irrigation Network Management Information System (INMIS), this SCADA integration allows water authorities to aggregate agricultural demand, calculate precise crop water requirements using sophisticated methodologies like the FAO Penman-Monteith equation, and execute rotational supply scheduling for truly equitable distribution.⁵² The ultimate triumph of the NLBC automation project was achieving its primary, historically elusive objective: successfully delivering designated, precisely measured water quotas to tail-end farmers for the first time in the history of the canal network, drastically improving water use efficiency and completely bypassing the losses associated with unmanaged earthen conveyance.⁵² Similar SCADA-based systems, such as those implemented by Tata Trusts in Nava Raipur, underscore that technology acts as the crucial mechanism to guarantee urban and agricultural water security.⁵⁵

Domain 04 of 04: Hydraulic Structures Engineering

Hydraulic structures—dams, barrages, spillways, canal falls, and aqueducts—represent the apex where civil engineering meets profound physical consequence. When these massive structures fail, they do not merely deteriorate; they release catastrophic kinetic energy capable

of leveling entire downstream communities. A staggering proportion of dam failures in India are not acts of unpredictable nature, but rather design events the structures were fundamentally unequipped to survive. Research into Indian dam failures attributes approximately 23% of structural collapses directly to inadequate spillway capacity, a figure exceeded only by foundation deficiencies, which are themselves often initiated by uncontrolled seepage and scour beneath weir floors under high-head conditions.⁵⁶ A structure that performs perfectly within its designed envelope offers absolutely no safety margin for the flood that exceeds it—and in India's changing climate, that flood is an inevitability.

The Consequence of Underestimated Hydrology: The Machchhu Dam Disaster

The historical benchmark for hydrological underestimation in the Indian subcontinent is the catastrophic failure of the Machchhu Dam II in Gujarat on August 11, 1979. Constructed as a composite structure spanning the Machhu River, it featured a masonry overflow spillway section 164.5 meters long, flanked by extensive earthen embankments extending thousands of meters on both sides.⁵⁸ The dam stood 74 feet above the riverbed, possessing a storage capacity of 1.1×10^8 cubic meters.⁵⁷ At the time of its commissioning in 1972, the dam was designed with a maximum spillway capacity of 200,000 cubic feet per second (cfs), a figure experts later agreed was fully compliant with the state of engineering practice for that period.⁵⁶

However, during a severe, anomalous monsoon event in August 1979, the peak inflow vastly exceeded all existing deterministic models. Post-failure forensic estimates of the peak inflow varied drastically, ranging from 460,000 cfs (according to the independent 1985 Murthy report) to an astonishing 630,000 to 936,000 cfs according to official government estimates.⁵⁶ Both figures represented an inflow more than double or triple the dam's maximum discharge capacity.

As the storm intensified on the night of August 10, operators attempted to manage the relentlessly rising water. By 1:30 AM on August 11, all functioning gates were fully opened (save for three that were malfunctioning due to a power failure that forced manual operation), pushing the spillway to pass 196,000 cfs—nearly its absolute maximum capacity.⁵⁶ Despite these efforts, the reservoir level continued to rise, eventually generating a 6.1-meter depth of water over the crest.⁵⁷ In the early afternoon, floodwaters completely overtopped the earthen embankments, cascading down the downstream slopes as a violently erosive cascade.⁵⁷ The subsequent catastrophic breaching of the dam, opening gaps from 20 to 200 meters wide, resulted in the loss of between 1,800 and 25,000 lives.⁵⁶ When the Machchhu Dam II was eventually rebuilt and completed in 1989, its spillway capacity was drastically and necessarily increased to 872,000 cfs, a grim acknowledgment of the original design's profound hydrological inadequacy.⁵⁶

Contemporary Failures: Construction Deficits and Mechanical Flaws

While historical failures often stemmed from a genuine lack of climatological data, modern hydraulic structure failures increasingly point to lapses in construction methodology, aging

infrastructure, and poor mechanical maintenance.

The near-collapse of the Karam Dam in the Dhar district of Madhya Pradesh in August 2022 serves as a critical case study in construction negligence. Designed as an earthfill dam to support irrigation, the structure suffered a catastrophic failure mechanism during its very first filling.⁶⁰ Forensic initial analysis indicated that the methodology adopted for the dam's construction deviated severely from its design and specifications.⁶¹ A critical mechanical error occurred in the concrete spillway portion: sluice valves proposed for the Non-Overflowing (N.O.F.) section were improperly installed.⁶¹ Consequently, mechanical failure under excess water pressure hindered the discharge of excess flow.⁶¹

Simultaneously, improper planning regarding nalla closure activities and a severe lack of compaction in the hearting zone—which comprised highly expansive black cotton soil—allowed severe piping and seepage to take hold.⁶¹ On the morning of August 12, a massive portion of the earthen bund, measuring approximately 10 meters in height and 20 meters in length, suffered a severe slope failure between RD 500m and 530m.⁶¹ The resulting deluge stripped adjacent tribal farming lands in Bhandakho of their topsoil, washing away the earth and leaving behind barren, rocky terrain that decimated local livelihoods.⁶³ The Karam incident underscores that even analytically sound hydraulic designs are instantly voided by substandard geotechnical execution and a lack of safety compliance.

Similarly, the catastrophic incident at the Dr. K.L. Rao Sagar Pulichintala Project on the Krishna River in Andhra Pradesh in August 2021 exposed the severe vulnerabilities of high-head mechanical systems. During a period of heavy upstream monsoon inflow, engineers attempted to lift crest gate number 16 to a height of 4 feet at 3:00 AM.⁶⁴ Due to a mechanical failure of the hydraulic lifting system under extreme pressure—exacerbated by a design flaw where the gap between gates in each pier was increased to 400 mm—the massive gate collapsed and was completely washed away.⁶⁴ The failure unleashed a massive, uncontrolled discharge exceeding 500,000 cusecs downstream toward the Prakasam Barrage.⁶⁵ Authorities were forced to immediately enact emergency stop-lock arrangements, inserting 11 heavy iron plates into the existing gate grooves to arrest the flow.⁶⁵ Such failures are frequently precipitated by a failure to maintain flood cushions, routine violation of operating rule curves, and a lack of rigorous pre-monsoon mechanical auditing.⁶⁶

Redefining Design Standards: Spillway Capacity and River Training

To mitigate the recurrence of such immense disasters, the Central Water Commission (CWC) and the Bureau of Indian Standards (BIS) continually revise the guidelines governing hydraulic analysis. The calculation of the Probable Maximum Flood (PMF)—the largest flood theoretically possible at a given location—remains the fundamental cornerstone of spillway design.⁶⁸ However, the engineering paradigm is steadily shifting from purely deterministic PMF calculations toward Risk-Informed Decision Making (RIDM).⁷⁰ This approach evaluates the probability of extreme events and correlates them with the socioeconomic consequences of structural failure, ensuring that spillways are sized not just for historical precedents, but for

modeled climatic extremes.⁷¹

The design of the spillway itself requires meticulous hydraulic profiling. CWC guidelines emphasize that velocities at the entrance of the spillway must be strictly limited, and the alignment of approach channel training walls must feature gradual transitions to minimize head loss and ensure uniform, non-turbulent approach flows.⁷² High-velocity flow over steep spillway chutes requires sophisticated modeling to assess aeration, shockwave generation, and cavitation risks, ensuring the concrete surface is not catastrophically eroded by negative vapor pressures under high-head conditions.⁷³ Furthermore, the rising popularity of orifice spillways—which serve the dual purpose of passing massive floods and flushing sediment—requires complex dimensional analysis of fluid properties, flow geometry, dynamic viscosity, and surface tension constraints.⁷⁴

Beyond dams and spillways, the interaction between linear infrastructure (such as highways and railways) and riverine systems requires equally rigorous hydraulic governance. The Indian Roads Congress standard IRC:89-2019 ("Guidelines for Design and Construction of River Training and Control Works for Road Bridges") dictates the stabilization parameters for bridges crossing major alluvial and meandering rivers like the Ganga, Yamuna, and Brahmaputra.⁷⁵ The construction of bridge piers inherently constricts the natural flow cross-section, generating an upstream afflux (a dangerous rise in water level) and localized increases in flow velocity.⁷⁵

To prevent outflanking and devastating foundation scour, IRC:89 and IS:10751 mandate the judicious design of guide bunds—earthen dykes heavily protected by stone pitching, crated stone-gabions on the sloping river-side face, and launching aprons at the toe that confine the river flow parallel to the abutments.⁷⁸ Additionally, transverse structures such as spurs or groynes are utilized to redirect high-velocity currents away from vulnerable banks.⁷⁵ The empirical design of these protective structures is highly reliant on 1D/2D hydraulic modeling (utilizing software such as HEC-RAS) to simulate backwater curves, bridge afflux, and accurately estimate pier scour depth.⁷⁹ An infrastructure crossing built without such predictive modeling generates immediate liability, threatening not only the bridge structure itself through foundation undermining but also adjacent agricultural and residential lands through artificially elevated upstream flood levels. Energy dissipators—including USBR Type I-IV stilling basins, trajectory buckets, and roller buckets—must be precisely proportioned at all canal falls and drop structures to ensure high-velocity tailwater does not erode downstream channel beds, causing secondary network failures.

By applying rigorous hydraulic analysis at every stage of structural design, engineering firms can ensure that no structure leaves the drafting table without a demonstrated, analytically proven hydraulic performance envelope, effectively securing both human life and capital investment.

Works cited

1. INVESTIGATING PREMATURE PAVEMENT FAILURE DUE TO MOISTURE Final

- Report - ROSA P, accessed May 11, 2026,
https://rosap.ntl.bts.gov/view/dot/22882/dot_22882_DS1.pdf
2. Examine the Underlying Causes of Flexible Pavement Deteriorations, Methods of Maintenance and Pavement Condition Rating, accessed May 11, 2026,
<https://www.ijeat.org/wp-content/uploads/papers/v10i4/D25180410421.pdf>
 3. Premature Pavement Distresses Study | PDF | Road Surface | Asphalt - Scribd, accessed May 11, 2026,
<https://www.scribd.com/document/396714841/Investigation-of-Premature-Pavement-Distresses-on-Typical-National-Highway-Project>
 4. Field experiment on the water immersion characteristics of graded ..., accessed May 11, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12511564/>
 5. MOISTURE VARIATION IN HIGHWAY SUBGRADES AND THE ASSOCIATED CHANGE IN SURFACE DEFLECTIONS - Transportation Research Board (TRB), accessed May 11, 2026,
<https://onlinepubs.trb.org/Onlinepubs/trr/1974/497/497-004.pdf>
 6. Subgrade Strength Recovery of Fine-Grained-Soil-Containing Roads in Western Oregon Forest - MDPI, accessed May 11, 2026,
<https://www.mdpi.com/1999-4907/15/4/671>
 7. PREMATURE FAILURE OF ASPHALT OVERLAYS FROM STRIPPING: CASE HISTORIES - Auburn University, accessed May 11, 2026,
<https://www.eng.auburn.edu/research/centers/ncat/files/technical-reports/rep01-01.pdf>
 8. Potential Impact of Climate Change on Rainfall Intensity-Duration-Frequency Curves in Roorkee, India - IDEAS/RePEc, accessed May 11, 2026,
https://ideas.repec.org/a/spr/waterr/v30y2016i13d10.1007_s11269-016-1441-4.html
 9. Impacts of Climate Change on IDF Relationships for Design of Urban Stormwater Systems, accessed May 11, 2026, <https://etd.iisc.ac.in/handle/2005/3037>
 10. Climate change impacts on rainfall intensity-duration-frequency curves in local scale catchments - PubMed, accessed May 11, 2026,
<https://pubmed.ncbi.nlm.nih.gov/38489074/>
 11. Comprehensive Overview of IRC Code: Guidelines on Road Drainage (IRC:SP:42-2014), accessed May 11, 2026,
<https://www.roadvision.ai/blog/comprehensive-overview-of-irc-code-guidelines-on-road-drainage-irc-sp-42-2014>
 12. IRC SP 42-2014 Guidelines On Road Drainage PDF - Scribd, accessed May 11, 2026,
<https://www.scribd.com/document/460917419/IRC-SP-42-2014-Guidelines-on-Road-Drainage-pdf>
 13. IRC SP 042: Guidelines on Road Drainage (First Revision) - Internet Archive, accessed May 11, 2026, <https://archive.org/details/govlawiracy2014sp42>
 14. DRAINAGE FOR QUALITY IMPROVEMENT OF RURAL ROADS S.K.Mazumder (Individual Consultant) Former AICTE Emeritus Professor Delhi Colle, accessed May 11, 2026,
https://www.profskmazumder.com/assets/img/conferencepapers/Con_Ref_61.pdf

15. GUIDELINES FOR ROAD DRAINAGE SYSTEM | MoIT, accessed May 11, 2026, <https://www.moit.gov.bt/wp-content/uploads/2022/05/Guideline-to-Road-Drainage-System.pdf>
16. Full text of "IRC SP 042: Guidelines on Road Drainage (First Revision)" - Internet Archive, accessed May 11, 2026, https://archive.org/stream/govlawiracy2014sp42/govlawiracy2014sp42_djvu.txt
17. Identifying urban floods based on hec-ras, swmm, and google earth engine rms, accessed May 11, 2026, <https://icsbe.uui.ac.id/wp-content/uploads/2024/02/Paper-D-31.pdf>
18. An Introduction to Green Cities and Comparing HEC-RAS and EPA SWMM: A Quick Guide from Clear Creek Solutions, accessed May 11, 2026, <https://www.clearcreeksolutions.info/blog/streamflowarticle-eafc9-9nyjz-9tx8b-n4z6m-yd98b>
19. (PDF) Application of SWMM for Urban Storm Water Management: A ..., accessed May 11, 2026, https://www.researchgate.net/publication/389704755_Application_of_SWMM_for_Urban_Storm_Water_Management_A_Case_Study_of_Hyderabad_City
20. Identification of Critical Hotspots in Urban Drainage Networks using MIKE URBAN, accessed May 11, 2026, <https://www.chijournal.org/H539>
21. Comparing SWMM and HEC-RAS Hydrological Modeling Performance in Semi-Urbanized Watershed - MDPI, accessed May 11, 2026, <https://www.mdpi.com/2073-4441/17/9/1331>
22. Dynamic Groundwater resources of India | Vikaspedia - Energy, accessed May 11, 2026, <https://energy.vikaspedia.in/viewcontent/energy/database/dynamic-groundwater-resources-of-india?lgn=en>
23. National Compilation on - Department of Water Resources, River Development and Ganga Rejuvenation, accessed May 11, 2026, <https://www.jalshakti-dowr.gov.in/static/uploads/2025/11/711d8d476bbf498cd90f8038e7371227.pdf>
24. Union Minister of Jal Shakti Releases Dynamic Ground Water ... - PIB, accessed May 11, 2026, <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2089039>
25. National Compilation on Dynamic Ground Water Resources of India 2023 - Dataset - CKAN, accessed May 11, 2026, <https://data.opencity.in/dataset/national-compilation-on-dynamic-ground-water-resources-of-india-2023>
26. Groundwater depletion in India: the influence of seasonal precipitation and land use land cover - PMC, accessed May 11, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12804263/>
27. GRACE Sees Groundwater Losses Around the World, accessed May 11, 2026, <https://grace.jpl.nasa.gov/resources/9/grace-sees-groundwater-losses-around-the-world/>
28. Full article: Advances in GRACE satellite studies on terrestrial water storage: a comprehensive review - Taylor & Francis, accessed May 11, 2026, <https://www.tandfonline.com/doi/full/10.1080/10106049.2025.2482706>

29. Publications - GRACE Tellus - NASA, accessed May 11, 2026,
<https://grace.jpl.nasa.gov/publications/>
30. A Case Study on Construction Dewatering-Induced Settlement Damage: Could This Have Been Avoided? - David Publishing, accessed May 11, 2026,
<https://www.davidpublisher.com/index.php/Home/Article/index?id=33973.html>
31. Numerical analysis of metro station pit dewatering and its influence - Frontiers, accessed May 11, 2026,
<https://www.frontiersin.org/journals/earth-science/articles/10.3389/feart.2022.1120772/full>
32. A Case Study on Construction Dewatering-Induced Settlement Damage: Could This Have Been Avoided? - David Publishing, accessed May 11, 2026,
<https://www.davidpublisher.com/Public/uploads/Contribute/5a4063319c18f.pdf>
33. Mumbai Metro Line 3 Construction Report | PDF | Tunnel - Scribd, accessed May 11, 2026,
<https://www.scribd.com/document/756489219/Construction-Methodology-R1-21-05>
34. Construction Dewatering for Underground Station in Urban Environment - ResearchGate, accessed May 11, 2026,
https://www.researchgate.net/publication/344303842_Construction_Dewatering_for_Underground_Station_in_Urban_Environment
35. Borewell drilling damages Mumbai Metro 3 tunnel near CSMT - The Times of India, accessed May 11, 2026,
<https://timesofindia.indiatimes.com/city/mumbai/borewell-drilling-damages-mumbai-metro-3-tunnel-near-csmt/articleshow/129660210.cms>
36. Construction Follow-up Design Services for Delhi Metro Phase III Contract CC-04 (TBM, Cross-Passages & Shafts) - Omikron Kappa, accessed May 11, 2026,
https://omikronkappa.gr/wp-content/uploads/Delhi_Metro.pdf
37. Environmental Impact Assessment Study for Najafgarh- Dhansa Bus Stand Corridor of Delhi Metro, accessed May 11, 2026,
https://backend.delhimetrorail.com/documents/56/EIAReporN-D Bs_standCorridor.pdf
38. Construction & Engineering Laws and Regulations Report 2025-2026 India - ICLG.com, accessed May 11, 2026,
<https://iclg.com/practice-areas/construction-and-engineering-law-laws-and-regulations/india>
39. Construction Disputes in India - Nishith Desai Associates, accessed May 11, 2026,
https://www.nishithdesai.com/fileadmin/user_upload/pdfs/Research_Papers/Construction-disputes-in-india.pdf
40. M/S. Complete Dewatering vs M/S. Integral Construction Co. (India) on 27 March, 2015, accessed May 11, 2026, <https://indiankanoon.org/doc/46727782/>
41. Managing construction risks in Asia-Pacific: India | White & Case LLP, accessed May 11, 2026,
<https://www.whitecase.com/insight-our-thinking/managing-construction-risks-asia-pacific-india>
42. Hydrospatial Modelling and Simulations for Assessing the Irrigation Canal

- Conveyance Losses - ResearchGate, accessed May 11, 2026,
https://www.researchgate.net/publication/332117702_Hydrospatial_Modelling_and_Simulations_for_Assessing_the_Irrigation_Canal_Conveyance_Losses
43. Irrigation in Central Asia in figures - Food and Agriculture Organization, accessed May 11, 2026, <https://www.fao.org/4/i3289e/i3289e.pdf>
 44. 9 VI June 2021 <https://doi.org/10.22214/ijraset.2021.35361>, accessed May 11, 2026, <https://www.ijraset.com/files/serve.php?FID=35361>
 45. The importance of irrigation supply chains within the water footprint: an example from the Pakistani part of the Indus basin, accessed May 11, 2026,
https://www.waterfootprint.org/resources/The_importance_of_irrigation_supply_chains_within_the_water_footprint_an_example_from_the_Pakistani_part_of_the_Indus_basin.pdf
 46. Irrigation water withdrawal - Food and Agriculture Organization, accessed May 11, 2026,
<https://www.fao.org/aquastat/en/data-analysis/irrig-water-use/irrig-water-withdrawal/>
 47. Emerging Issues and Problems of Soil Salinity and Water Logging: A Case Study of Indira Gandhi Canal, Rajasthan - ResearchGate, accessed May 11, 2026,
https://www.researchgate.net/publication/369959145_Emerging_Issues_and_Problems_of_Soil_Salinity_and_Water_Loading_A_Case_Study_of_Indira_Gandhi_Canal_Rajasthan
 48. Emerging Issues and Problems of Soil Salinity and Water Logging: A Case Study of Indira Gandhi Canal, Rajasthan - Knowledgeable Research, accessed May 11, 2026,
<https://knowledgeableresearch.com/index.php/1/article/download/416/357/695>
 49. Study Of Waterlogging And Soil Salinization Problem In And - ijrass, accessed May 11, 2026, https://www.ijrass.com/2017/Vol_4-Issue_4/paper_26.pdf
 50. Problem of soil salinity and waterlogging in Indira Gandhi canal area of Rajasthan state., accessed May 11, 2026,
<https://www.cabidigitallibrary.org/doi/full/10.5555/19971911521>
 51. POLICY PAPAER 49 COVER.cdr - AgEcon Search, accessed May 11, 2026,
<https://ageconsearch.umn.edu/record/358870/files/PP49.pdf>
 52. Canal Automation for Smart Digital Irrigation Management: A Case ..., accessed May 11, 2026,
https://www.researchgate.net/publication/336148668_Canal_Automation_for_Smart_Digital_Irrigation_Management_A_Case_Study_on_Narayanpur_Left_Bank_Canal_Karnataka_State_India
 53. AGENDA, accessed May 11, 2026,
https://icid-ciid.org/icid_data_web/Webinar-MedhaRubicon301025.pdf
 54. scada-iot based irrigation modernisation under pmksy - PIB, accessed May 11, 2026,
<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2238841@=3&lang=1>
 55. Designing Systems for Water Security - Tata Trusts, accessed May 11, 2026,
https://www.tatatrusts.org/Upload/Content_Files/case-study-nava-raipur-SCADA.pdf

56. Machhu Dam II (Gujarat, India, 1979) | ASDSO Dam Failures and ..., accessed May 11, 2026,
<https://damfailures.org/case-study/machhu-dam-ii-gujarat-india-1979>
57. 41.2 History of Dam Failures - Nptel, accessed May 11, 2026,
https://archive.nptel.ac.in/content/storage2/courses/105106114/pdfs/Unit41/41_2.pdf
58. 1979 Machchhu dam failure - Wikipedia, accessed May 11, 2026,
https://en.wikipedia.org/wiki/1979_Machchhu_dam_failure
59. Dam Failure Case Study: Machhu Dam II (Gujarat, India, 1979), accessed May 11, 2026,
<https://damsafety.org/reference/dam-failure-case-study-machhu-dam-ii-gujarat-india-1979>
60. Karam Dam Failure: Causes and Impact | PDF - Scribd, accessed May 11, 2026,
<https://www.scribd.com/presentation/690238172/CDworks>
61. Documentation on Dam Leakage, Breaching and Disaster ... - NIDM, accessed May 11, 2026,
<https://nidm.gov.in/PDF/pubs/Karam%20Dam%20NIDM%202023.pdf>
62. Karam Dam Failure: Lessons Learned | PDF | Dam | Water And The Environment - Scribd, accessed May 11, 2026,
<https://www.scribd.com/document/856021614/Failure-of-Karam-dam-Case-Study-Lesson-Learnt-Vivek-Tripathi-CE-CWC>
63. Ground Report: Karam Dam Deluge Strips Tribal Farmers of Land, Leaving Rocky Soil Crisis, accessed May 11, 2026,
<https://en.themooknayak.com/tribal-news/ground-report-karam-dam-deluge-strips-tribal-farmers-of-land-leaving-rocky-soil-crisis>
64. Pulichintala project hydraulic gate washes away due to heavy discharge - The Hindu, accessed May 11, 2026,
<https://www.thehindu.com/news/national/andhra-pradesh/pulichintala-project-hydraulic-gate-washes-away-due-to-heavy-discharge/article35736666.ece>
65. High alert as Pulichintala project gate collapses - The New Indian Express, accessed May 11, 2026,
<https://www.newindianexpress.com/states/andhra-pradesh/2021/Aug/06/high-alert-as-pchintala-project-gate-collapses-2340969.html>
66. Pulichintala project's fallen gate fixed with stop lock; Opposition TDP blames YSRCP government's poor maintenance - Deccan Herald, accessed May 11, 2026,
<https://www.deccanherald.com/india/pulichintala-projects-fallen-gate-fixed-with-stop-lock-opposition-tdp-blames-ysrcp-governments-poor-maintenance-1017345.html>
67. Andhra Pradesh: Pulichintala Dam Gate disaster shows improper maintenance and operation - SANDRP, accessed May 11, 2026,
<https://sandrp.in/2021/08/12/andhra-pradesh-pulichintala-dam-gate-disaster-shows-improper-maintenance-and-operation/>
68. Estimating the Probable Maximum Flood (PMF) Starting with No Existing Model, accessed May 11, 2026,
<https://www.hec.usace.army.mil/confluence/hmsdocs/hmsguides/hms-modeling-ap>

[lications/probable-maximum-flood-tutorials/estimating-the-probable-maximum-flood-pmf-starting-with-no-existing-model](#)

69. FOR Oroville PMF memo draft - California Water Commission, accessed May 11, 2026,
https://cwc.ca.gov/-/media/CWC-Website/Files/Documents/2023/09_September/September2023_Item_8_FriendsOfTheRiverComment.pdf
70. Modernizing Probable Maximum Precipitation Estimation (2024) - Earth & Environmental Systems Modeling, accessed May 11, 2026,
https://eesm.science.energy.gov/system/files/publications/attachments/Paciorek%202024%20highlight_NASEM_CASCADE%20%281%29_0.pdf
71. Estimating probable maximum flood probabilities - Association of State Dam Safety Officials, accessed May 11, 2026,
<https://damsafety.org/content/estimating-probable-maximum-flood-probabilities>
72. Compendium on Spillways and Energy Dissipators Designs - A Legacy of CWPRS, accessed May 11, 2026,
https://cwprs.gov.in/storage/pdf-uploads/Compendium_on_spillways_and_energy_dissipators_design-A_legacy_of_CWPRS.pdf
73. Spillways, Shockwaves and Air Entrainment - Review and recommendations, accessed May 11, 2026,
<https://damsafety.org/content/spillways-shockwaves-and-air-entrainment-review-and-recommendations>
74. TECHNICAL MEMORANDUM ON GUIDELINES FOR HYDRAULIC DESIGN OF ORIFICE SPILLWAY by Dr. (Mrs.) V. V. Bhosekar, Scientist E Dr. (Mrs - CWPRS, accessed May 11, 2026,
<https://cwprs.gov.in/storage/pdf-uploads/6%20%20Guidelines%20for%20Hydraulic%20Design%20of%20Orifice%20Spillway,%202017.pdf>
75. Understanding IRC Code 89: Guidelines for River Training and Control Works for Road Bridges - RoadVision AI, accessed May 11, 2026,
<https://www.roadvision.ai/blog/understanding-irc-code-89-guidelines-for-river-training-and-control-works-for-road-bridges>
76. IRC 89-1997: River Training Guidelines | PDF - Scribd, accessed May 11, 2026,
<https://www.scribd.com/document/433515346/IRC-089-1997>
77. IRC 089: Guidelines for Design and Construction of River Training and Control Works for Road Bridges (First Revision) - Internet Archive, accessed May 11, 2026,
https://archive.org/details/govlawircy1997sp89_0
78. RIVER TRAINING WORKS, accessed May 11, 2026,
https://www.profskmazumder.com/assets/img/journalpapers/Jrnl_Ref_41.pdf
79. Dam Break Analysis using HEC-RAS and Flood Inundation Modelling for Pulichinatala Dam in Andhra Pradesh, India - ResearchGate, accessed May 11, 2026,
https://www.researchgate.net/publication/350995254_Dam_Break_Analysis_using_HEC-RAS_and_Flood_Inundation_Modelling_for_Pulichinatala_Dam_in_Andhra_Pradesh_India